

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & M_{xxz} \\ -M_{xxx} & 0 & M_{xxz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & -M_{xxx} & 0 \end{bmatrix}$$

$$M_{xxx} = \mathbb{M}_{3a}^{(1)}$$

$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)}$$

$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & 0 \\ 0 & -\frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{M}_z^{(2)}$$

$$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ -M_{yzy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzy} = \mathbb{M}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & 0 \\ 0 & 0 & 0 & -M_{xyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xyz} & -M_{xyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyz} \\ 0 & 0 & 0 & 0 & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = \mathbb{M}_{4a}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & M_{xxy} \\ 0 & 0 & 0 & -M_{xyz} & 0 & -M_{xxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xyz} & M_{xyz} & 0 & 0 & M_{yzz} & 0 \\ 0 & 0 & 0 & -M_{yzz} & 0 & -M_{xyz} \\ -M_{xxy} & M_{xxy} & 0 & 0 & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = 2\mathbb{T}_{3a}^{(1)}$$

$$M_{xxy} = 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)}$$

$$M_{yzz} = -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzz} & 0 \\ -M_{yzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzz} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xyz} & 0 & M_{xxy} \\ -M_{xyz} & 0 & M_{xxy} \\ 0 & 0 & M_{zzx} \\ 0 & M_{yzz} & 0 \\ M_{yzz} & 0 & 0 \\ 0 & -M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = \mathbb{T}_{3a}^{(2)}$$

$$M_{xxy} = \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{zzx} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$

$$M_{yzz} = \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzz} & -M_{yzz} & 0 & 0 & M_{yzz} & 0 \\ 0 & 0 & 0 & M_{yzz} & 0 & -M_{yzz} \\ M_{xyx} & M_{xyx} & M_{xyz} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzz} = \mathbb{T}_{3a}^{(3)}$$

$$M_{yzz} = \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & 0 & M_{xxy} & M_{xzy} & 0 & M_{xyx} \\ 0 & 0 & 0 & -M_{xyz} & 0 & -M_{xxy} & -M_{xzy} & 0 & -M_{xxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzx} & 0 & 0 & -M_{zzx} \\ M_{yzz} & -M_{yzz} & 0 & 0 & M_{yzz} & 0 & 0 & M_{yzz} & 0 \\ 0 & 0 & 0 & -M_{yzz} & 0 & M_{yzz} & -M_{yzz} & 0 & M_{yzz} \\ -\frac{M_{xxy}}{2} - \frac{M_{xzy}}{2} & \frac{M_{xxy}}{2} + \frac{M_{xzy}}{2} & 0 & 0 & M_{xzy} & 0 & 0 & M_{xyz} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxyz} &= \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} + \mathbb{T}_{3a}^{(2)} \\
M_{xxxy} &= 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\
M_{xxzy} &= \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} - \mathbb{T}_{3a}^{(2)} \\
M_{xxyx} &= 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} \\
M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\
M_{yzxx} &= \mathbb{M}_{4a}^{(1)} - 2\mathbb{T}_{3a}^{(1)} \\
M_{yzzx} &= -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\
M_{yzzz} &= -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
M_{yzxx} & -M_{yzxx} & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzzx} & 0 \\
0 & 0 & 0 & M_{yzzx} & 0 & -M_{yzxx} & M_{yzzx} & 0 & -M_{yzxx} \\
M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{yzxx} &= \mathbb{T}_{3a}^{(3)} \\
M_{yzzx} &= -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\
M_{yzzz} &= \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyxx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyzz} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & 0 & 0 \\
0 & 0 & M_{yyyy} \\
0 & 0 & 0 \\
0 & M_{yyyy} & 0 \\
0 & 0 & 0 \\
0 & 0 & -M_{yyyy} \\
0 & 0 & 0 \\
0 & -M_{yyyy} & 0 \\
0 & 0 & 0 \\
-M_{yyyy} & 0 & 0
\end{bmatrix}$$

$$M_{yyyy} = -\mathbb{M}_{4a}^{(1)}$$